

SIMATS ENGINEERING**Slot: C, Course Code: ITA0501, Course Name: Computer Vision****Duration: 60 mins****Marks: 20****Activity Tool 3: Class Test (Low)****Assessment Tool Weightage: 10%****COURSE OUTCOME COVERED****CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)**

Q. No.	Question
1	An image appears too bright. Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.
2	A dark image lacks detail. Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.
3	An image has poor contrast. Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.
4	A low-contrast image requires uniform intensity distribution. Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.
5	An image shows uneven brightness across regions. Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.
6	A noisy image contains random intensity variations. Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.
7	An image requires noise reduction without significant loss of details. Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.
8	An image appears blurred after smoothing. Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.
9	Fine details in an image are not clearly visible. Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.
10	A noisy image must be preprocessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

Code of Conduct:

I Aratipalli Sathwika Reddy (192525274) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

A. Sathwika Reddy

Signature of the Student
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RUBRICS FOR CALCULATION

Criteria	<i>Understanding of Concepts</i>	<i>Explanation</i>	<i>Application</i>	<i>Organization</i>	<i>Accuracy</i>	<i>Clarity</i>	Total
Weightage	25%	25%	20%	15%	10%	5%	100%

Signature of the Course Faculty

1) Concept :-

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A very bright image has most pixel values concentrated near the higher intensity range (255). This causes loss of details in bright regions.

Suitable Technique

Negative (inverse) Grey Level Transformation

Formula

$$S = L - 1 - r$$

where S = output pixel intensity

r = input pixel intensity

$L = 256$ for an 8-bit image

Application Process

1. Read the image
2. Apply the negative transformation to every pixel.
3. Replace each pixel using the above formula.

Justification

- * The negative transformation effectively reduces excessive brightness by reversing the gray levels.
- * Bright areas become darker, making hidden structures and fine details easier to observe.
- * This method is especially useful for images where excessive illumination, reduces visual clarity, such as X-ray images, scanned documents, and Satellite images.

Expected outcome

- * Brighter image
- * Improved shadow details
- * Better visibility
- * Enhanced low-intensity regions

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Concept :-

Dark images contain pixel values mainly in the low-intensity range, making objects difficult to identify.

Suitable Technique

Log Transformation

$$S = c \log(1+r) \quad [c \text{ is constant}]$$

Application Process

1. Read the dark image.
2. Apply logarithmic transformation.
3. Normalize pixel values.
4. Display the enhanced image.

Justification:- Log transformation expands low intensity values while compressing high-intensity values.

Expected Outcomes :-

- * Brighter image
- * Improved shadow details
- * Better visibility

3) **Concept** :- poor contrast occurs when pixel values occupy only small portion of the intensity range.

Suitable Technique: Histogram Equalization

Working Principle:-

Histogram Equalization redistributes pixel intensities over the entire grayscale range.

Steps

1. Compute histogram
2. Calculate cumulative distribution function.
3. Normalise CDF
4. Map old intensities to new values
5. Generate enhanced image.

Expected Outcome

- * Higher contrast
- * Better edge visibility
- * Improved object detection
- * Enhanced overall image quality.

44 Concept :-

when intensity values are clustered, contrast decreases.

Suitable Method

Histogram Equalization

working process

- * Compute histogram
- * Obtain cumulative histogram
- * Normalise cumulative values.
- * Replace original intensities with equalized values.

Effect on intensity values

pixel values become uniformly distributed across 0-255

Expected outcome:-

- * The image become sharper and visually clearer.

5) Concept :-

Uneven illumination causes some areas to appear brighter while others remain darker.

Suitable Technique

Adaptive Histogram Equalization

Working process

1. Dividing image into small blocks
2. perform histogram equalization separately.
3. Limit contrast amplification
4. Merge all blocks smoothly.

Justification:-

CLAHE enhances local contrast without over-amplifying noise.

Expected outcome

- * uniform brightness
- * Better local contrast
- * Improved visibility.

68 Concept :-

Noise causes unwanted intensity fluctuations that reduces image quality.

Suitable Filter

Median Filter

Working Mechanism

- * Select a window (3×3)
- * Arrange neighbouring pixels in ascending order.
- * Replace the center pixel with median value.

Justification

unlike averaging filters, the median filter removes impulse noise while preserving edges.

Expected outcome

- * Noise reduction
- * Edge preservation
- * Better image quality
- * Improved processing

7. Concept :-

The enhancement method should remove noise while preserving image edges

Suitable Technique

Bilateral Filter

Working process

The filter considers:

- * Spatial distance

- * pixel intensity difference

only similar neighboring pixels contribute while preserving important edges.

Expected outcome:

- * Effective noise removal

- * Sharp edges retained

- * Improved visual quality

- * Better feature extraction

54 Concept

Smoothing removes noise but also reduces edge sharpness.

Suitable Technique

Laplacian sharpening

$$g(x, y) = f(x, y) - \nabla^2 f(x, y)$$

Working principle

1. Compute Laplacian
2. Detect high-frequency components
3. Add edge information back to image

Expected outcome

- * Sharper edges
- * Improved detail
- * Better object visibility
- * Enhanced image clarity

9) Concept :-

From details corresponding to high-frequency
Suitable Technique

High pass filter
working principle

High-pass filters suppress low-frequency components

Expected outcome

- * Enhanced edges
- * Clear-texture

10) Concept :-

Spatial filtering modifies pixel value
Suitable method - Gaussian filter

Expected outcome -

- * Reduced random noise
- * Smooth image
- * Better preprocessing